

Energy Through Trophic Levels: Teacher Key

1. One-hectare solar ledger

Quoted solar constant:

$$S = 1361 \text{ W m}^{-2}.$$

$$P = SA = 1361 \times 10,000 = 1.361 \times 10^7 \text{ W} = 13.61 \text{ MW}.$$

For one hour,

$$E = Pt = (1.361 \times 10^7)(3600) = 4.8996 \times 10^{10} \text{ J} = 48,996 \text{ MJ} = 48.996 \text{ GJ}.$$

This is an idealised top-of-atmosphere/normal-incidence opening budget; real ground-level input is lower and variable.

Stage	Pass %	Waste %	In / MJ	Passed / MJ	Wasted here / MJ	Cum. waste / MJ
Sun → grass	1.8	98.2	48996	881.928	48114.072	48114.072
Grass → rabbit	12	88	881.928	105.83136	776.09664	48890.16864
Rabbit → fox	9	91	105.83136	9.5248224	96.3065376	48986.4751776

Overall pass fraction:

$$0.018 \times 0.12 \times 0.09 = 0.0001944,$$

so final pass-through = 0.01944% and cumulative waste = 99.98056%.

Energy cross-check:

$$9.5248224 + 48986.4751776 = 48996 \text{ MJ}.$$

Percentage cross-check:

$$0.01944\% + 99.98056\% = 100\%.$$

2. Three trophic ledgers

A Pond: waste percentages 97.5%, 82%, 86%, 93%. Passed energies: 375 kJ; 67.5 kJ; 9.45 kJ; 0.6615 kJ. Stage wastes: 14625 kJ; 307.5 kJ; 58.05 kJ; 8.7885 kJ. Final pass-through = 0.00441%; cumulative waste = 99.99559%. Check:

$$0.6615 + 14999.3385 = 15000 \text{ kJ}.$$

B Crop direct: waste percentages 97.8% and 75%. Crop energy = 220 kJ; human energy = 55 kJ. Final pass-through = 0.55%; cumulative waste = 99.45%.

C Woodland: waste percentages 98.8%, 85%, 89%, 92%. Passed energies: 360 kJ; 54 kJ; 5.94 kJ; 0.4752 kJ. Final pass-through = 0.001584%; cumulative waste = 99.998416%. Check:

$$0.4752 + 29999.5248 = 30000 \text{ kJ}.$$

3. Multiplicative discovery

Adding gives 22.8%, which does not agree. Decimal efficiencies are 0.018, 0.12, 0.09.

$$0.018 \times 0.12 \times 0.09 = 0.0001944.$$

Thus

$$\boxed{\eta_{\text{total}} = \eta_1 \eta_2 \eta_3 \cdots \eta_n}.$$

The reason is that every later fraction acts on the energy remaining after the previous transfer.

4. Waste as an independent accounting route

For pass efficiencies 20% then 30%:

$$0.20 \times 0.30 = 0.06 = 6\% \text{ final pass-through,}$$

so cumulative waste is 94%.

Starting with 1000 kJ: stage 1 passes 200 kJ and wastes 800 kJ. Stage 2 passes 60 kJ and wastes 140 kJ. Therefore

$$60 + 800 + 140 = 1000 \text{ kJ.}$$

The stage waste percentages cannot be added because the 70% applies only to the 200 kJ entering stage 2, not to the original 1000 kJ.

5. Direct calculations

Grass $\rightarrow$ vole $\rightarrow$ owl:

$$0.08 \times 0.13 = 0.0104 = 1.04\% \text{ pass; } \quad 98.96\% \text{ cumulative waste.}$$

Phytoplankton $\rightarrow$ krill $\rightarrow$ penguin $\rightarrow$ seal:

$$0.16 \times 0.11 \times 0.09 = 0.001584 = 0.1584\% \text{ pass; } \quad 99.8416\% \text{ waste.}$$

Maize $\rightarrow$ insect $\rightarrow$ chicken $\rightarrow$ human:

$$0.07 \times 0.18 \times 0.12 = 0.001512 = 0.1512\% \text{ pass; } \quad 99.8488\% \text{ waste.}$$

6. Reverse ledger

0.12% = 0.0012. If the third pass efficiency is x ,

$$0.10 \times 0.15 \times x = 0.0012 \quad \Rightarrow \quad x = 0.08 = 8\%.$$

Corresponding stage waste = 92%. Overall cumulative waste = $100\% - 0.12\% = 99.88\%$.

7. Interpretation

The crop-to-human pathway has the greatest final pass-through, 0.55%. Shortening a chain removes one or more multiplicative bottlenecks. A statement such as “90% is wasted at each level” refers to 90% of the energy entering that particular level, not 90% of the original starting account each time.

Close the books: multiplication; 100%.